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Monitored Anesthetic Care Combined with Scalp Nerve Block in Awake Craniotomy: An Effective Attempt at Enhanced Recovery After Neurosurgery

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■ BACKGROUND: **Enhanced recovery after surgery has been attempted in neurosurgery at a greater rate. However, concern exists regarding the feasibility of using enhanced recovery after neurosurgery (ERANS). How to manage available resources to safely perform ERANS and improve clinical outcomes has been the subject of much debate and discussion.**

■ METHODS: **Owing to the paucity of data available on the use of ERANS protocols, we performed the present feasibility study. We studied the outcomes of the protocols used within a tertiary referral neurosurgery center. Data from patients who had undergone awake craniotomy within an ERANS protocol were prospectively recorded in our institution from September 2017 to December 2018. We also evaluated the safety and effectiveness of the novel ERANS protocol.**

■ RESULTS: **A total of 20 patients (mean age, 49.5 ± 17.8 years) were included in the present study. Intraoperative hypertension, hypotension, and bradycardia were present in 4 (20%), 1 (5%), and 1 (5%) patient, respectively. The postoperative morbidities included epilepsy in 1 (5%), pain in 3 (15%), and nausea or vomiting in 2 (10%). No significant changes had occurred in the mean arterial pressure, heart rate, blood glucose, or lactic acid level throughout the procedure. The median length of intensive care unit stay and postoperative hospital stay were 1 and 9.5 days, respectively. No 30-day readmissions or reoperations occurred during the present study.**

■ CONCLUSIONS: **Applying an ERANS protocol was feasible, associated with a low incidence of complications, and acceptable intensive care unit and postoperative hospital lengths of stay. The findings from the present study might provide a new approach for the further research of ERANS.**

INTRODUCTION

Neurosurgical procedures have traditionally been associated with long hospital lengths of stay (LOSs), long rehabilitation periods, and high costs. Thus, many patients in low-income countries have been unable to access potentially lifesaving neurosurgical procedures.[1] An enhanced recovery after surgery (ERAS) protocol was first proposed by Kehlet et al in 1997, with further reports in 2001 and 2002 by Wilmore and Kehlet.[2,3] ERAS is an evidence-based multimodal perioperative program that focuses on reducing stress, promoting functional recovery, and reducing medical costs.[2,3] At present, ERAS is considered appropriate for 70% of surgical patients, including patients undergoing colorectal surgery, urological

Key words
- Awake craniotomy
- Enhanced recovery after neurosurgery
- Monitored anesthetic care
- Scalp nerve block

Abbreviations and Acronyms
AC: Awake craniotomy
ATN: Auriculotemporal nerve
CLC: Compound lidocaine cream
ERAS: Enhanced recovery after surgery
ERANS: Enhanced recovery after neurosurgery
GDFT: Goal-directed fluid therapy
GON: Greater occipital nerve
LON: Lesser occipital nerve
LOS: Length of stay
MAC: Monitored anesthetic care
OCIM: Obliquus capitis inferior muscle
POD: Postoperative day
ScCM: Semispinalis capitis muscle
SNB: Scalp nerve block
SON: Supraorbital nerve

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surgery, orthopedic surgery, cardiac surgery and gynecological surgery.[4,5] Recently, several studies have developed and implemented ERAS protocols for patients undergoing neurosurgery.[6-8] However, all these protocols have mainly focused on perioperative patient care. In the present study, we have described a novel enhanced recovery after neurosurgery (ERANS) protocol, in addition to the regular ERAS interventions. We have emphasized some unique ideas such as shortened preoperative fasting, goal-directed fluid therapy (GDFT), intraoperative consumption of limited water, improving intraoperative monitoring strategies, no indwelling catheter, the use of monitored anesthetic care (MAC) combined with an ultrasound-guided scalp nerve block (SNB) instead of general anesthesia for neurosurgery, the early resumption of an oral diet and exercise, and appropriate discharge criteria.

METHODS

Design and Setting

In the present prospective study, we collected data from patients who had undergone elective awake craniotomy (AC) during a 16-month period from September 2017 to December 2018 at The First Affiliated Hospital, Jinan University. All the operations were performed or supervised by the same surgical and anesthesia team. All the patients and their families provided written informed consent, and the ethics committee of the hospital approved the present study.

Data Collection

The prospectively collected data included demographic details and baseline comorbidities. Data on the use of anesthetics, intraoperative hemodynamics, blood gas analysis results, intraoperative and postoperative complications, and LOS were extracted retrospectively from the medical records. The LOS for all the patients is presented as the median and the other data as the mean ± standard deviation.

ERANS Protocol

In accordance with the principles of multimodal ERAS pathways used for patients undergoing other elective procedures, the ERANS protocol (Figure 1) used in the present study included preoperative, intraoperative, and postoperative interventions.

Preoperative interventions

- Extensive perioperative counseling
- Patients abstain from both alcohol and smoking for ≥2 weeks
- Antimicrobial prophylaxis
- Shortened fluid fasts

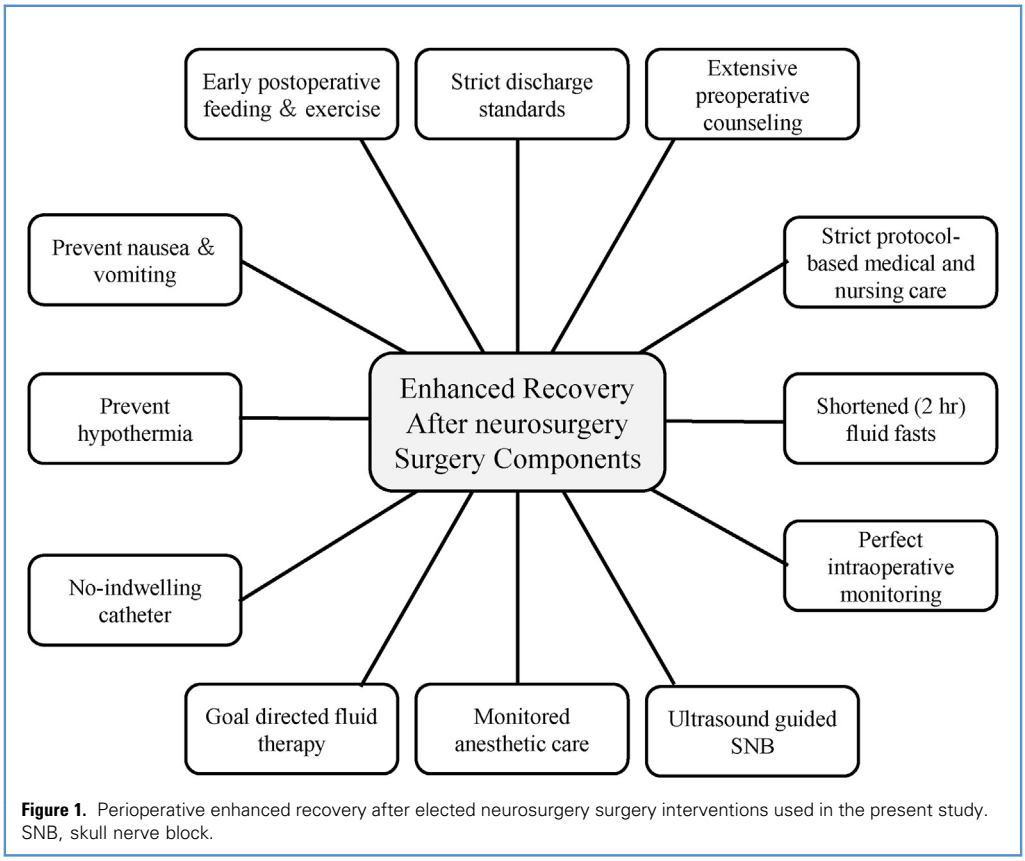


Figure 1. Perioperative enhanced recovery after elected neurosurgery surgery interventions used in the present study. SNB, skull nerve block.



Figure 2. All participants wore adult diapers instead of indwelling catheters during surgery.

Patient consumption of oral carbohydrates 2 hours before surgery

Patients practice wearing diapers 24 hours before surgery (Figure 2)

The use of 5% compound lidocaine cream (CLC) to relieve the pain of nerve blocks and vascular punctures (Figure 3)

Adjustment of the room temperature to 25°C before the patient enters the operating room

Intraoperative interventions

Intraoperative monitoring indexes



Figure 3. Compound lidocaine cream, 5%, was applied to the radial artery puncture point (1.5 g/cm^2) and covered with a transparent dressing.

Monitoring, including electrocardiography, noninvasive blood pressure measurements, body temperature, end-tidal carbon dioxide, and invasive blood pressure

The use of the bispectral index monitoring system to guide the requirement for intraoperative anesthetics

Blood gas analysis was performed once each hour to better assess oxygen delivery to tissue and oxygen consumption of tissue

Ultrasound-guided SNB[9-11]

Supraorbital nerve (SON)

For SON blockade, the patients were placed in a supine position and the probe was positioned above and parallel to the eyebrow. Cross scanning was performed by gently moving the probe toward the eyebrow.[12] As the supraorbital notch was revealed on the ultrasound screen, a 50-mm-long, 20-guage, short bevel needle was inserted. After the location of the needle tip was confirmed, 1 mL of 0.4% ropivacaine was infused slowly after negative aspiration of blood (Figure 4).

Auriculotemporal nerve (ATN)

For ATN blockade, the patients were placed in a supine position with the head turned to the contralateral side. After the zygomatic arch was identified, the probe was placed parallel and above the zygomatic arch to locate the superficial temporal artery. Next, a 50-mm-long, 20-guage, short bevel needle was inserted posterior to the superficial temporal artery, and 2 mL of 0.4% ropivacaine was slowly injected after negative aspiration of blood (Figure 5).

Greater occipital nerve (GON)

For GON blockade, the patients were placed in the prone position. After confirming the spinous process of C2, the probe was placed parallel to the laminae of C2. Next, we moved the probe slightly to confirm the semispinalis capitis muscle (ScCM) and obliquus capitis inferior muscle (OCIM). A reliable and consistent relationship exists between the GON and ScCM and OCIM. Thus, the ScCM and OCIM were used as the primary landmarks to locate the GON.[10,11] Once the GON was located, a 50-mm-long, 20-guage, short bevel needle was inserted to the GON and 4 mL of 0.4% ropivacaine was injected around the GON after negative aspiration of blood (Figure 6).

Lesser occipital nerve (LON)

For ultrasound-guided LON blockade, the patients were positioned supine with the head turned to the contralateral side. Using a transverse ultrasound scan and sliding the probe along the sternocleidomastoid muscle, the LON can be identified at the posterolateral border of the sternocleidomastoid muscle near where the spinal nerve emerges. Once the LON location was confirmed, the block was performed with a 50-mm-long, 20-guage, short bevel needle, and 2 mL of 0.4% ropivacaine was injected after negative aspiration of blood (Figure 7).

The procedures were repeated on the contralateral side



Figure 4. Position of the ultrasound probe during supraorbital nerve block and ultrasound image demonstrating needle placement. Ultrasound image showing that the tip of the needle (*white arrow*) has reached the supraorbital notch via the out-plane technique.

All ultrasound-guided injections were performed using a high-frequency linear ultrasound probe (6–14 Hz [Wisonic Medical Technology Co., Guangdong, China])
In-plane technology was used for all the SNBs, except for the SON blockade

MAC

All the patients had received a continuous infusion of propofol, dexmedetomidine, and sufentanil to maintain the bispectral index value between 60 and 80

Equipment for emergency airway control should be available throughout the procedure, including intubation under direct vision, blind nasal intubation, fiberoptic-assisted intubation, and/or laryngeal mask airway insertion

Their use should be planned for before beginning the procedure

Liquid management

Although the effectiveness of GDFT for ERAS is controversial, we still incorporated hemodynamics-guided intraoperative GDFT in our ERANS protocol

We advocate proper restrictive intravenous rehydration during the operation

As a supplement, we also provide appropriate oral rehydration when the patient complains of thirst

Other intraoperative interventions

All the patients were given palonosetron 0.15 mg and dexamethasone 10 mg before anesthesia induction to prevent nausea and vomiting

Haloperidol (1 mg) was given, if necessary

An inflatable heated blanket and fluid-heating device should be used in every case to prevent intraoperative and postoperative hypothermia.

The use of diapers instead of an indwelling catheter can reduce patient discomfort and avoid possible urinary tract infections

Postoperative interventions

Active or passive exercise was encouraged from postoperative day (POD) 1 to prevent the loss of muscle mass and the development of venous thrombosis

We also encourage enteral nutrition as early as possible

Discharge standards

The absence of dizziness, headache, nausea, vomiting, fever, and other symptoms

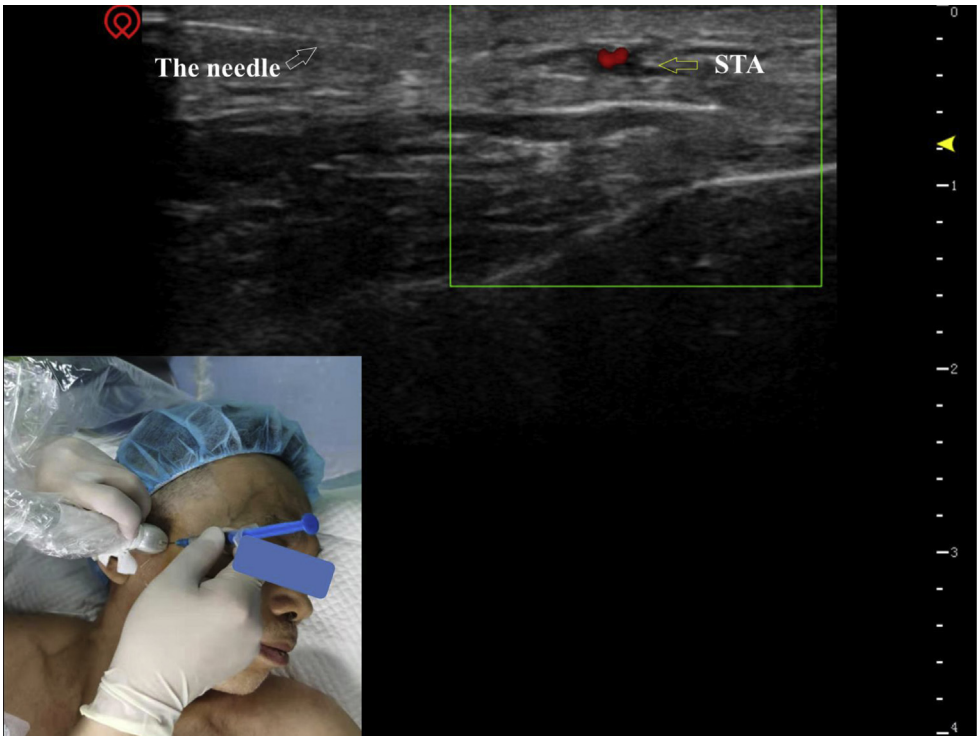


Figure 5. Position of the ultrasound probe during auriculotemporal nerve block and ultrasound image demonstrating needle placement. Using color Doppler ultrasound imaging, the superficial temporal artery (STA) was easily identified in the plane. The auriculotemporal nerve was seen as hypoechoic area close to the artery.

A well-healed incision

Imaging findings showing no intracranial space-occupying hemorrhage or edema

No obvious abnormalities found in the laboratory test results

RESULTS

During the study period, 20 patients had undergone elective AC with MAC combined with SNB. None of the patients enrolled in the ERANS pathway were subsequently excluded operatively or postoperatively. The baseline demographic and comorbidity data are shown in Table 1.

All the neurosurgical procedures were successfully completed. Most of the patients had presented with common neurological diseases such as meningioma, glioma, trigeminal neuralgia. The relevant details of the anesthesia and surgery are presented in Table 2. The mean heart rate and mean arterial pressure at different intervals are depicted in Figure 8. Documented oxygen desaturation (<90%) occurred in 1 patient and was corrected immediately after the patient had awoken. The blood glucose and lactic acid levels were slightly elevated, with no statistically significant differences (Figure 9).

The postoperative complications and LOS are presented in Table 3. For postoperative dietary management, all the patients started water intake immediately after surgery. All the patients received a polymeric nutritional supplement drink and solid foods at 4 and 24 hours after surgery, respectively. No patient died in-hospital.

DISCUSSION

Preoperative Management of ERANS Protocol

The normal preoperative interventions of ERAS were strictly implemented in our ERANS protocol. We also included some special preoperative interventions in our protocol. These included extensive perioperative counseling, the use of CLC, and the use of diapers instead of an indwelling urinary catheter.

First, a detailed step-by-step overview of our ERANS protocol was provided to all patients before surgery, because their anxiety can be minimized by explanations of the procedure. We reassure the patients that in the unlikely event of them finding the awake aspect of the surgery intolerable they can be reanesthetized. We found that most of our patients did not find the procedure distressing.

Second, evidence has shown that local dermal application of the CLC can be used to manage the pain associated with epidermal

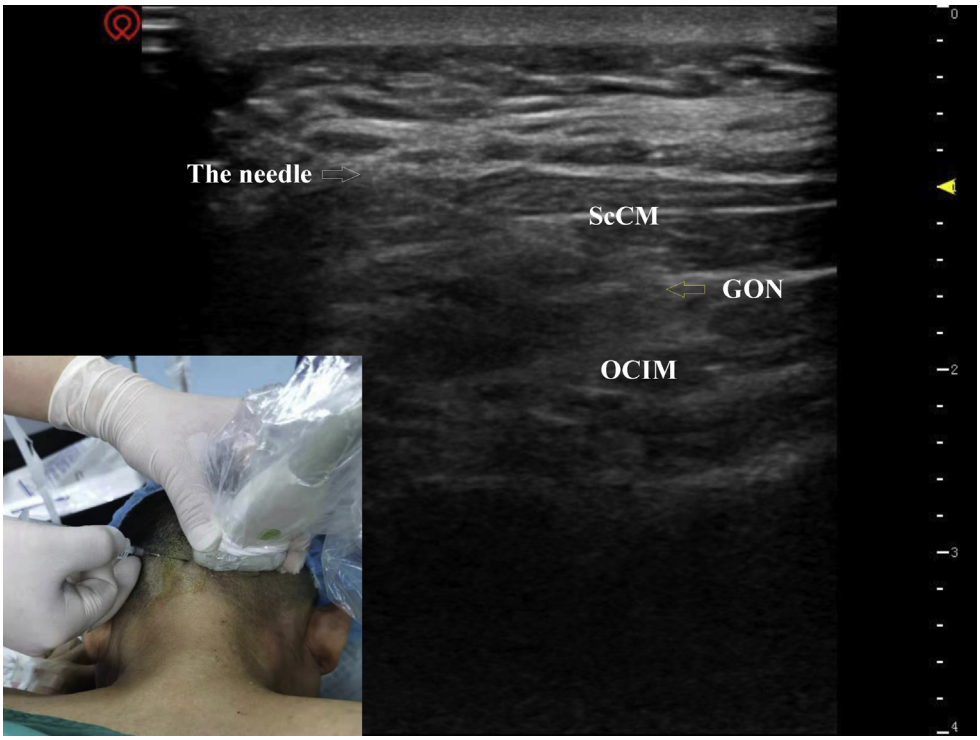


Figure 6. Position of the ultrasound probe during the greater occipital nerve (GON) block and ultrasound image demonstrating needle placement. The GON (*yellow arrow*) was typically identified lying superficial to the obliquus capitis inferior muscle (OCIM). ScCM, semispinalis capitis muscle.

injury.[13] In our protocol, CLC was applied to the puncture areas 30 minutes before the patient entered the operating room to relieve the pain caused by the punctures.

Third, Because the presence of an indwelling catheter makes patients feel unwell and increases the risk of urinary tract infections, previous ERAS protocols have suggest that early bladder catheter removal should be promoted to avoid the increased risk of infection and promote mobility. In our protocol, we have encouraged the patients to wear diapers instead of having an indwelling catheter placed. Because of the unusual urination habits of adults, patients should practice urinating in diapers for 1 day before surgery. No postoperative urinary retention and no catheter-related infections occurred in any of the patients.

Intraoperative Management of ERANS Protocol

Historically, AC was promoted for the surgical treatment of epilepsy in particular. At present, it is widely used for the resection of functional area lesions because it avoids the complications of general anesthesia and has been associated with shorter LOS and fewer complications and deaths.[14,15] Because of these benefits, our protocol has identified AC as the primary intervention. SNB, MAC, airway management, and experience in the specific technique used are required to successfully perform AC.[16]

The scalp is mainly innervated by the SON, ATN, GON, and LON.[17] The use of local anesthesia is the cornerstone of AC and has been established as the reference standard for patients undergoing AC,[18] including local infiltration and SNB. The research has shown that although both local infiltration and SNB can achieve satisfactory results, SNB can be performed with less medication and provides a longer anesthetic duration, meeting the needs of longer surgery.[19] Traditional SNB is performed with anatomical localization. To achieve better results, repeated punctures will often be required, resulting in additional injury to the patients. In contrast, ultrasound-guided SNB has the advantages of providing perfect blocking effects, less injury, and fewer complications. Thus, we included the use of the SNB in our protocol. The results showed that all the patients in the present study had experienced good analgesic effects, and no patient had complained of intraoperative pain. In addition,

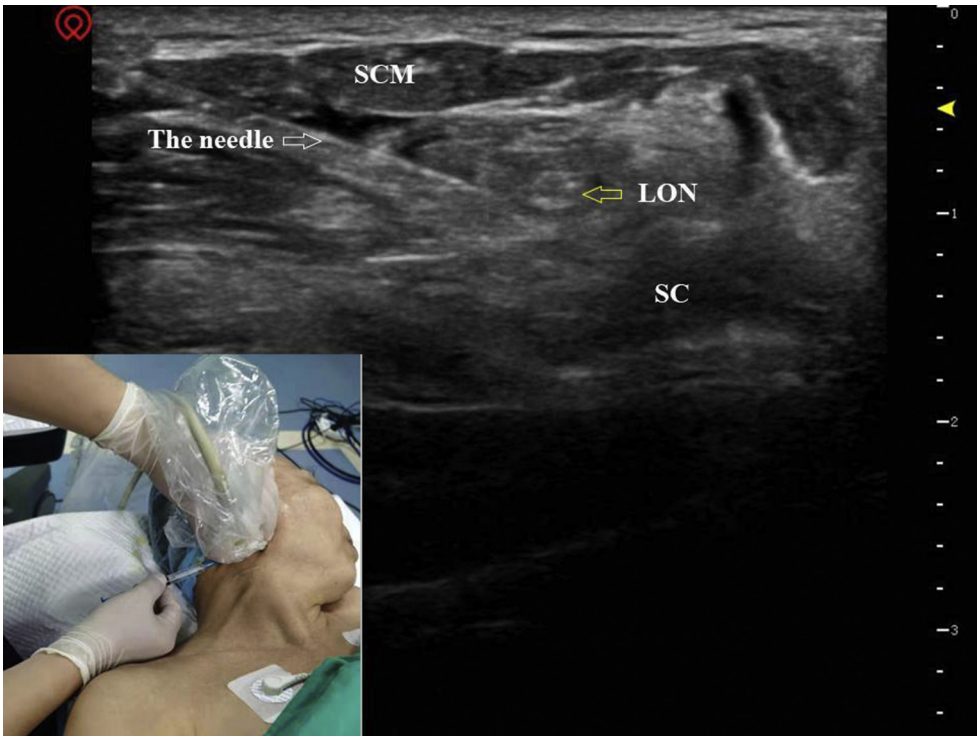


Figure 7. Position of the ultrasound probe and ultrasound image demonstrating needle placement. The lesser occipital nerve (LON; *yellow arrow*) emerges from underneath the posterior border of the sternocleidomastoid muscle (SCM). SC, splenius capitis.

ultrasound-guided SNB reduces the requirement for intravenous anesthetics, thereby reducing the incidence of anesthetic-related complications.

A safe and effective sedation strategy is the key to successful implementation of AC. In our protocol, we obtained adequate sedation with the simultaneous infusion of propofol, dexmedetomidine, and sufentanil. Propofol has been the most often used sedation agent for AC. It has the advantages of rapid onset and antiemetic and antiepileptic properties.[20,21] Dexmedetomidine is a selective α_2 receptor agonist and has been proposed as the main sedative and hypnotic. It produces a dose-dependent sedative effect and has the effects of relieving anxiety and pain and reducing cerebral blood flow. At present, dexmedetomidine is widely used in AC because of these properties.[22,23] Perhaps because its use can make it difficult for patients to wake up during surgery, sufentanil is rarely used in AC. However, in our protocol, a small dose of sufentanil was used to allow the patients to be more comfortable during the operation. Our results have shown that all the patients could wake up easily and communicate clearly with the doctor. Respiratory depression and airway obstruction are the inherent risks associated with MAC. Severe respiratory depression can lead to hypercapnia and brain swelling. A series of measures, including a lateral position, close monitoring, and a reduction in the use of opioids, were applied to prevent the occurrence of respiratory depression in our protocol. Equipment for emergency airway control should be available throughout the procedure. In the present study, only one patient developed transient hypoxemia, which it was corrected after the patient had awakened.

Evidence has suggested that limiting perioperative intravenous fluid administration can reduce the LOS and the incidence of postoperative complications and can also provide the neurosurgeon with an optimized field, although it can lead to a relative deficit.[7,24] To meet the dual goals of hemodynamic stability and a good surgical field, a hemodynamically guided GDFT was adopted in our ERANS protocol. Possibly resulting from our use of minimally invasive craniotomy, adequate analgesia and sedation, and appropriate oral rehydration, no significant hemodynamic fluctuations occurred in any of our 20 patients. The brief and slight decrease in mean arterial

Table 1. Baseline Demographic and Comorbidity Data ($n = 20$)

Variable	Mean ± SD or *n* (%)
Age (years)	49.5 ± 17.8
Sex	
Male	10 (50)
Female	10 (50)
BMI (kg/m^2)	
<18.5	0 (0)
18.5–24.9	17 (85)
>25	3 (15)
ASA classification	
I	16 (80)
II	4 (20)
Concomitant disease	
Hypertension	2 (10)
Diabetes mellitus	2 (10)
Moderate anemia	1 (5)
Smoker	3 (15)
Alcohol use	1 (5)

SD, standard deviation; BMI, body mass index; ASA, American Society of Anesthesiologists.

pressure at the beginning of surgery could have resulted from hypersecretions.

Patients undergoing craniotomy will frequently experience nausea and vomiting because of the surgical stimulation, including manipulation of the dura, temporal lobe, and meningeal vessels.[25] In our protocol, palonosetron 0.15 mg and dexamethasone 10 mg were used routinely to prevent nausea and vomiting.[26] In addition, the use of our protocol diminished the need for intraoperative opioid supplementation during surgery, which can also reduce the incidence of nausea and vomiting. Two patients had experienced very mild intraoperative nausea, which was relieved by the additional administration of haloperidol. Intraoperative hypothermia can lead to cardiac events, bleeding, wound infections, and pressure ulcers.[27] Several randomized trials and meta-analyses have shown that mild hypothermia is associated with adverse outcomes in patients undergoing noncardiac operations.[28,29] In our ERANS protocol, measures such as maintaining an appropriate operating room temperature and the use of an inflatable heating blanket and fluid heating device were applied to prevent intraoperative hypothermia. The results showed that no patients experienced hypothermia in the present study. The blood glucose and lactic acid levels have been considered to be closely related to the prognosis of patients.[30-32] In our protocol, a range of useful measures, including adequate preoperative counseling, mask oxygen inhalation, and sufficient sedation and analgesia, were taken to avoid intraoperative hyperglycemia and hyperlactic acidemia. The results showed that a slight increase had occurred in both the blood glucose and the lactic acid. However, the differences from the preoperative baseline to the levels during surgery were not statistically significant.

Postoperative Management in the ERANS Protocol

Postoperative pain has been a common patient complaint after neurosurgery. Opioids, nonopioid analgesia, and local anesthetic infusions have usually been used to manage postoperative pain in recent ERANS protocols. In our protocol, SNB was the major strategy for intra- and postoperative analgesia. As a supplement, nonsteroidal anti-inflammatory drugs were given to patients to help relieve postoperative pain. Three patients in our series had complained of postoperative pain and required nonsteroidal anti-inflammatory drugs. Postoperative opioids were not routinely used

Table 2. Details and Outcomes of Surgery and Anesthesia

Variable	*n* (%) or Mean ± SD
Diagnosis	
Meningioma	4 (20)
Glioma	7 (35)
Trigeminal neuralgia	6 (30)
Acoustic neuroma	1 (5)
Hemangioblastoma	1 (5)
Epidermoid cyst	1 (5)
Surgery	
Patient position during procedure	
Right lateral	16 (80)
Left lateral	4 (20)
Procedure duration (minutes)	334.9 ± 22.8
Blood loss (mL)	241.7 ± 39.7
Anesthesia	
Propofol consumption (mg)	358.3 ± 38.0
Dexmedetomidine consumption (μg)	116.9 ± 13.0
Sufentanil consumption (μg)	26.9 ± 2.2
Intraoperative crystalloid (mL)	894.4 ± 90.8
Intraoperative colloid (mL)	350 ± 99.8
Red blood cell transfusion	1 (2 U)
Blood plasma transfusion	1 (400 mL)
Transient intraoperative morbidity	
Hypoxemia	0 (0)
Hypertension	4 (20)
Hypotension	1 (5)
Bradycardia	1 (5)
Tachycardia	0 (0)

SD, standard deviation.

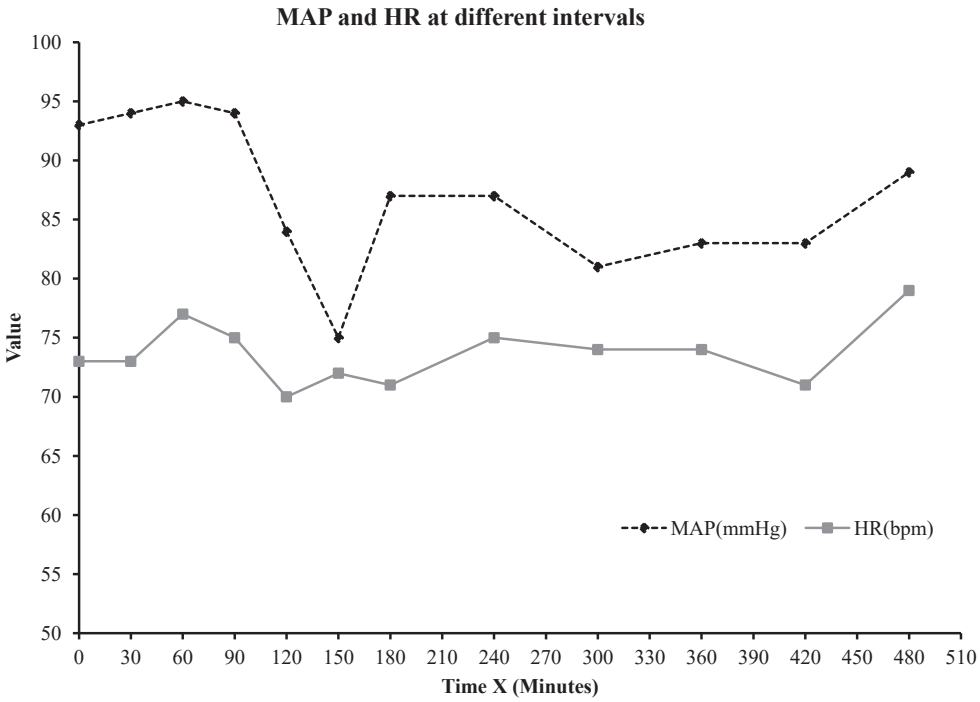


Figure 8. Changes in mean arterial pressure (MAP) and heart rate (HR) at different time points. No significant changes occurred in either the MAP or the HR during the entire surgery ($P > 0.05$).

in our protocol because of their limited efficacy and extensive side effects.

The early resumption of an oral diet and exercise is a traditional standard ERAS component.[33-35] With to the minimal usage of opioids and, thus, lower effects on gastrointestinal function, we have encourage all patients to resume their normal oral diet as early as possible. In our protocol, oral water intake was started immediately after surgery (and even during surgery, if necessary), and an oral polymeric nutritional supplement drink and solid food intake were started 4 hours after surgery and on POD 1, respectively. It is well-known that bed rest is associated with several deleterious effects, such as muscle atrophy, deep vein thrombosis, and insulin resistance. In our protocol, all the patients started exercise from POD 1. Those patients who lose motor function should engage in passive movement exercises. In the present study, no bed rest–related complications occurred.

The LOS is the key indicator for the whole study. According to other reports, compared with general anesthesia, patients undergoing AC will have a shorter LOS.[36,37] Although we did not perform an analysis of randomized controlled trials, a retrospective analysis of other neurosurgical cases at our institution revealed that patients undergoing the ERANS protocol had shorter intensive care unit LOSs and acceptable postoperative hospital LOSs.

The present study had some limitations. First, we lacked a control group with which to compare the outcomes from the ERANS pathway. Second, postoperative LOS was used as the primary endpoint of the present study. However, we did not perform an economic analysis of the effects of using the ERANS protocol. Finally, a sampling bias could have been present because of the small sample size and single-institution series.

CONCLUSIONS

The results from the present study have demonstrated that applying the novel ERANS protocol is feasible, safe, and associated with low morbidity and acceptable intensive care unit and postoperative

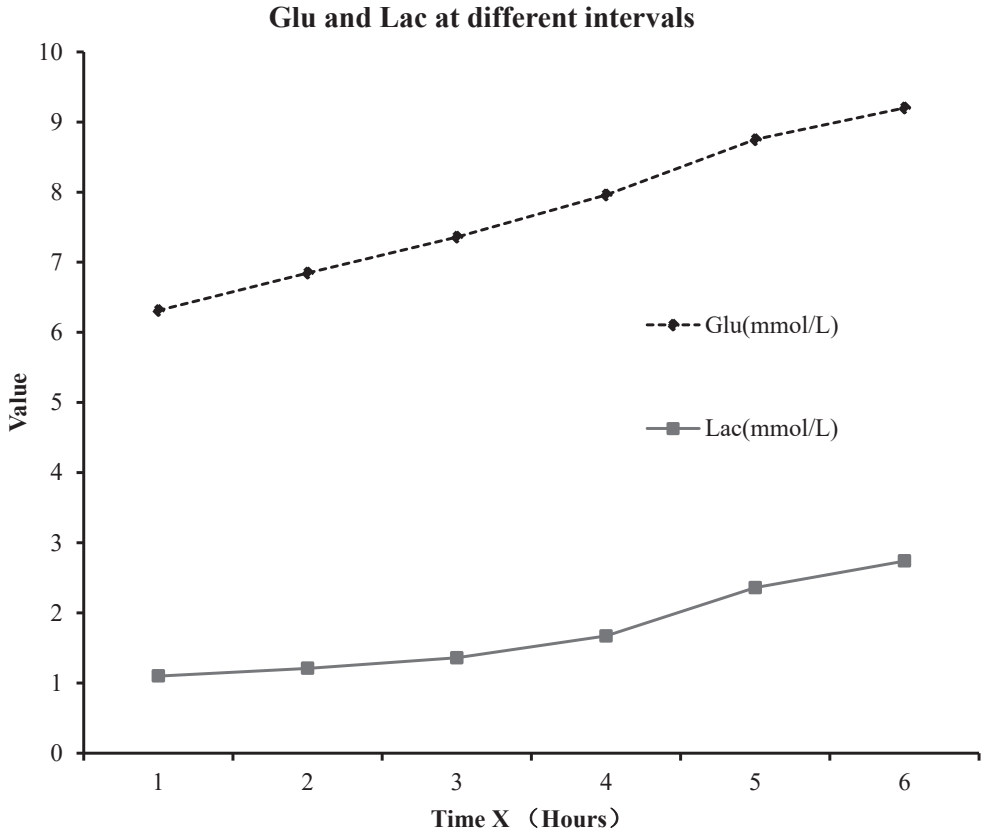


Figure 9. Changes in glucose (Glu) and lactic acid (Lac) at different time points. Measurements were performed every 60 minutes. A slight increase occurred in both glucose and lactic acid, but the differences from baseline were not statistically significant ($P > 0.05$).

Table 3. Postoperative Morbidity and Outcome Measures

Variable	*n* (%)
Postoperative morbidity	
Respiratory complications	0 (0)
Cardiovascular complications	0 (0)
Thromboembolic complications	0 (0)
Postoperative bleeding	0 (0)
Surgical site or intracranial infection	0 (0)
Epilepsy	1 (5)
Catheter-associated infection	0 (0)
Median ICU LOS stay (days)	1 (5)
Median postoperative hospital LOS (days)	9.5
Reoperation ≤30 days	0 (0)
Readmission ≤30 days	0 (0)

ICU, intensive care unit; LOS, length of stay.

hospital LOSs. The findings from the present study could provide a new approach for the further research of ERANS protocols.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

Yan-jun Chen: Conceptualization, Methodology, performing clinical procedures, Formal analysis, Writing — original draft, Writing — review & editing. **Cai Nie:** Conceptualization, Methodology, Formal analysis, Data curation, Writing — original draft, Writing — review & editing. **Hao Lu:** Performing clinical procedures, Funding acquisition, Formal analysis, Writing — review & editing. **Liu Zhang:** Performing clinical procedures, Writing — review & editing. **Hong-lin Chen:** Ultrasound-guided nerve block, Writing — review & editing. **Shi-yong Wang:** Performing clinical procedures, Writing — review & editing. **Wei Li:** Conceptualization, Methodology, Writing — review & editing. **Si Shen:** Conceptualization, Methodology, Ultrasound-guided nerve block, Writing — original draft, Final approval of the version to be published, Writing — review & editing. **Hao Wang:** Conceptualization, Methodology, Revising manuscript critically for important intellectual content, Final approval of the version to be published, Writing — review & editing.

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